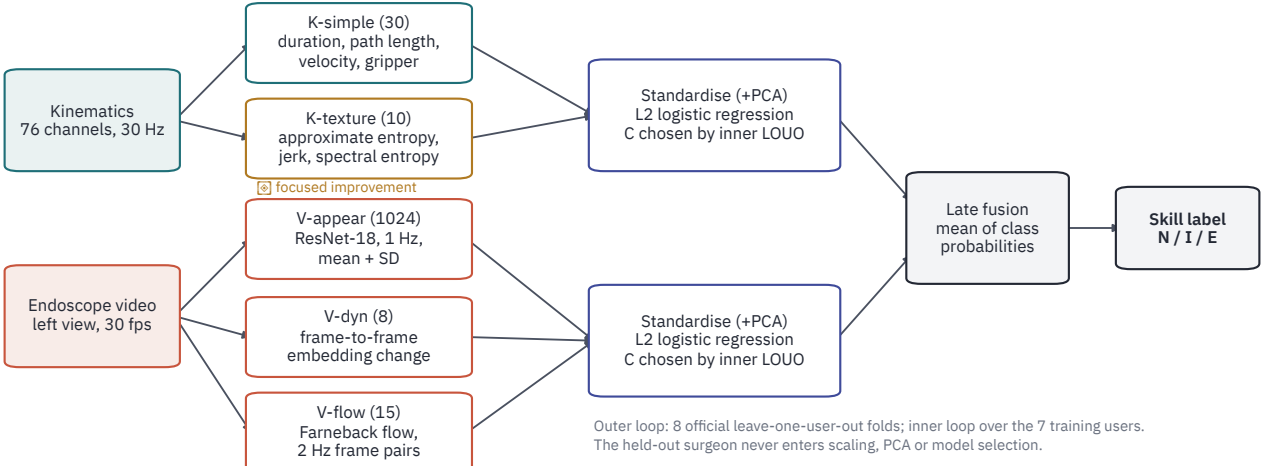
Kinematics
76 channels, 30 Hz
K-simple (30)
duration, path length,
velocity, gripper
K-texture (10)
approximate entropy,
jerk, spectral entropy
focused improvement
Endoscope video
left view, 30 fps
V-appear (1024)
ResNet-18, 1 Hz,
mean + SD
V-dyn (8)
frame-to-frame
embedding change
V-flow (15)
Farneback flow,
2 Hz frame pairs
Standardise (+PCA)
L2 logistic regression
C chosen by inner LOUO
Standardise (+PCA)
L2 logistic regression
C chosen by inner LOUO
Late fusion
mean of class
probabilities
Skill label
N / I / E
Outer loop: 8 official leave-one-user-out folds; inner loop over the 7 training users.
The held-out surgeon never enters scaling, PCA or model selection.